Ketron® 1000 PEEK

Poly-ether-ether-ketone

Ketron® 1000 Polyetheretherketone PEEK is an unfilled, general purpose grade that offers the highest elongation and toughness of all materials in the PEEK family. Ideal for instrument and seal components, where ductility and inertness are critical, Ketron® 1000 PEEK shapes are known for their ability to fit within a variety of applications and industries.

PRODUCT DATASHEET

		ISO*			ASTM*		
		Test methods	Units	Indicative values	Test methods	Units	Indicative values
Thermal properties (1)	Melting temperature (DSC, 10°C (50°F) / min)	ISO 11357-1/-3	°C	340	ASTM D3418	°F	644
	Glass transition temperature (DMA- Tan δ) (2)		°C			°F	
	Thermal conductivity at 23°C (73°F)		W/(K.m)	0.25		BTU in./(hr.ft².°F)	1.75
	Coefficient of linear thermal expansion (-40 to 150 °C) (-40 to 300°F)				ASTM E-831 (TMA)	µin./in./°F	26
	Coefficient of linear thermal expansion (23 to 60°C) (73°F to 140°F)		µm/(m.K)				
	Coefficient of linear thermal expansion (23 to 100°C) (73°F to 210°F)		µm/(m.K)	50			
	Heat Deflection Temperature: method A: 1.8 MPa (264 PSI)	ISO 75-1/-2	°C	160	ASTM D648	°F	320
	Continuous allowable service temperature in air (20.000 hrs) (3)		°C	250		°F	480
	Min. service temperature (4)		°C	-50		°F	
	Flammability: UL 94 (3 mm (1/8 in.)) (5)			V-0			V-0
	Flammability: Oxygen Index	ISO 4589-1/-2	%	35			
Mechanical Properties (6)	Tensile strength	ISO 527-1/-2 (7)	MPa	115	ASTM D638 (8)	PSI	16,000
	Tensile strain (elongation) at yield	ISO 527-1/-2 (7)	%	5	ASTM D638 (8)	%	5
	Tensile strain (elongation) at break	ISO 527-1/-2 (7)	%	17	ASTM D638 (8)	%	40
	Tensile modulus of elasticity	ISO 527-1/-2 (9)	MPa	4300	ASTM D638 (8)	KSI	630
	Shear Strength			55	ASTM D732	PSI	8000
	Compressive stress at 1 / 2 / 5 % nominal strain	ISO 604 (10)	MPa	38 / 75 / 140			
	Compressive strength				ASTM D695 (11)	PSI	20,000
	Charpy impact strength - unnotched	ISO 179-1/1eU	kJ/m²	no break			
	Charpy impact strength - notched	ISO 179-1/1eA	kJ/m²	3.5			
	Izod Impact notched				ASTM D256	ft.lb./in	0.6
	Flexural strength	ISO 178 (12)	MPa	170	ASTM D790 (13)	PSI	25,000
	Flexural modulus of elasticity	ISO 178 (12)	MPa		ASTM D790	KSI	600
	Rockwell M hardness (14)	ISO 2039-2		105	ASTM D785		100
	Shore Hardness D (14)	ISO 868		85	ASTM D2240		85
Electrical Properties	Electric strength	IEC 60243-1 (15)	kV/mm	24	ASTM D149	Volts/mil	480
	Volume resistivity	IEC 62631-3-1	Ohm.cm	10^14	ASTM D257	Ohm.cm	
	Surface resistivity	ANSI/ESD STM 11.11	Ohm	10^13	ANSI/ESD STM 11.11	Ohm	10^13
	Dielectric constant at 1 MHz	IEC 62631-2-1		3.2	ASTM D150		3.3
	Dissipation factor at 1MHz	IEC 62631-2-1		0.002	ASTM D150		0.003
Miscellaneous	Colour			Natural (beige), Black, Red			Natural (beige), Black, Red
	Density	ISO 1183-1	g/cm³	1.31			
	Specific Gravity				ASTM D792		1.31
	Water absorption after 24h immersion in water of 23 °C (73°F)	ISO 62 (16)	%	0.06	ASTM D570 (17)	%	0.10
	Water absorption at saturation in water of 23 °C (73°F)		%	0.45	ASTM D570 (17)	%	0.5
	Wear rate	ISO 7148-2 (18)	µm/km	28	QTM 55010 (19)	In³.min/ft.lbs.hrX10-¹⁰	375
	Dynamic Coefficient of Friction (-)	ISO 7148-2 (18)		0.3-0.5	QTM 55007 (20)		0.32
	Limiting PV at 100 FPM (safety factor 4)				QTM 55007 (21)	ft.lbs/in².min	8500
	Limiting PV at 0.1 / 1 m/s cylindrical sleeve bearings		MPa.m/s	0.33 / 0.21			
	Chemical Resistance	www.mcam.com/en/support/chemical-resistance-information			www.mcam.com/en/support/chemical-resistance-information		

Note: 1 g/cm³ = 1,000 kg/m³ ; 1 MPa = 1 N/mm² ; 1 kV/mm = 1 MV/m

NYP: there is no yield point

This table, mainly to be used for comparison purposes, is a valuable help in the choice of a material. The data listed here fall within the normal range of product properties of dry material. However, they are not guaranteed and they should not be used to establish material specification limits nor used alone as the basis of design. See the remaining notes on the next page.

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Notes, see datasheet on page 1

1. The figures given for these properties are for the most part derived from raw material supplier data and other publications.
2. Values for this property are only given here for amorphous materials and for materials that do not show a melting temperature (PBI & PI).
3. Temperature resistance over a period of min. 20,000 hours. After this period of time, there is a decrease in tensile strength – measured at 23 °C – of about 50 % as compared with the original value. The temperature value given here is thus based on the thermal-oxidative degradation which takes place and causes a reduction in properties. Note, however, that the maximum allowable service temperature depends in many cases essentially on the duration and the magnitude of the mechanical stresses to which the material is subjected.
4. Impact strength decreasing with decreasing temperature, the minimum allowable service temperature is practically mainly determined by the extent to which the material is subjected to impact. The value given here is based on unfavourable impact conditions and may consequently not be considered as being the absolute practical limit.
5. These estimated ratings, derived from raw material supplier data and other publications, are not intended to reflect hazards presented by the material under actual fire conditions. There is no ‘UL File Number’ available for these stock shapes.
6. Most of the figures given for the mechanical properties are average values of tests run on dry test specimens machined out of end products with a wall thickness thicker than required by the specific test standard. All tests are done at room temperature (23°C / 73°F)
7. Test speed: either 5 mm/min or 50 mm/min [chosen acc. to ISO 10350-1 as a function of the ductile behaviour of the material (tough or brittle)] using type 1B tensile bars
8. Test speed: either 0.2"/min or 2"/min [chosen as a function of the ductile behaviour of the material (brittle or tough)] using Type 1 tensile bars
9. Test speed: 1 mm/min, using type 1B tensile bars
10. Test specimens: cylinders Ø 8 mm x 16 mm, test speed 1 mm/min
11. Test specimens: cylinders Ø 0.5" x 1", or square 0.5" x 1", test speed 0.05"/min
12. Test specimens: bars 4 mm (thickness) x 10 mm x 80 mm ; test speed: 2 mm/min ; span: 64 mm.
13. Test specimens: bars 0.25" (thickness) x 0.5" x 5" ; test speed: 0.11"/min ; span: 4"
14. Measured on 10 mm, 0.4" thick test specimens.
15. Electrode configuration: Æ 25 / Æ 75 mm coaxial cylinders ; in transformer oil according to IEC 60296 ; 1 mm thick test specimens.
16. Measured on discs Ø 50 mm x 3 mm.
17. Measured on 1/8" thick x 2" diameter or square
18. Test procedure similar to Test Method A: “Pin-on-disk” as described in ISO 7148-2, Load 3MPa, sliding velocity= 0,33 m/s, mating plate steel Ra= 0.7-0.9 µm, tested at 23°C, 50%RH.
19. Test using journal bearing system, 200 hrs, 118 ft/min, 42 PSI, steel shaft roughness 16±2 RMS micro inches with Hardness Brinell of 180-200
20. Test using Plastic Thrust Washer rotating against steel, 20 ft/min and 250 PSI, Stationary steel washer roughness 16±2 RMS micro inches with Rockwell C 20-24
21. Test using Plastic Thrust Washer rotating against steel, Step by step increase pressure, test ends when plastic begins to deform or if temperature increases, depending on the material, to a maximum which lays between 212°F (100°C) and 482°F (250°C), a 4:1 safety factor has been applied to the posted value.

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